

Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Program:

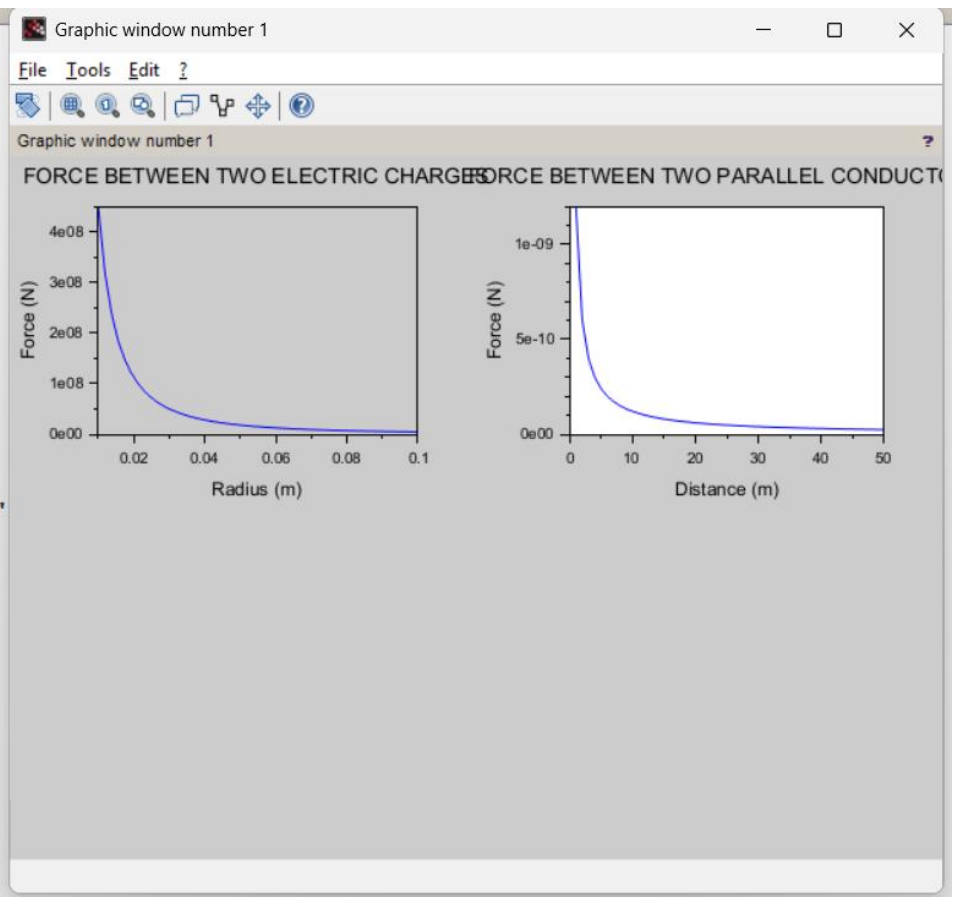
```
1  clc;
2  clear;
3  charge_1 = input("Enter the charge_1 value .-");
4  charge_2 = input("Enter the charge_2 value .-");
5  radius = input("Enter the radius value (m) .-");
6  current_1 = input("Enter the current_1 value .-");
7  current_2 = input("Enter the current_2 value .-");
8  l = input("Enter the length value (m) .-");
9  d = input("Enter the distance value (m) .-");
10 epsilon = 8.854e-12;
11 mu = 4 * %pi * 1e-7;
12 F = (charge_1 * charge_2) ./ (4 * %pi * epsilon * radius^2)
    ;
13 f = (mu * current_1 * current_2 * l) ./ (2 * %pi * d);
14 disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES");
15 disp(F);
16 disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS");
17 disp(f);
18 x = linspace(0.01, 0.1, 50);
19 y = (charge_1 * charge_2) ./ (4 * %pi * epsilon * x.^2);
20 z = (mu * current_1 * current_2 * l) ./ (2 * %pi * (1:50))
    ;
21 figure();
22 subplot(2,2,1);
23 plot(x, y);
24 xlabel("Radius (m)");
25 ylabel("Force (N)");
26 title("FORCE BETWEEN TWO ELECTRIC CHARGES");
27 subplot(2,2,2);
28 plot(1:50, z);
29 xlabel("Distance (m)");
30 ylabel("Force (N)");
31 title("FORCE BETWEEN TWO PARALLEL CONDUCTORS");
32
```

Output:

```
Enter the charge_1 value = 3**10^-6
Enter the charge_2 value = 5*10^-6
Enter the radius value (m) = 0.03
Enter the current_1 value = 2
Enter the current_2 value = 3
Enter the length value (m) = 0.001
Enter the distance value (m) = 10

"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
49931957.
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
1.200D-10

-->
```



Calculation:

Exp:- 3

$$\text{Charge 1} = 3 \times 10^{-6}$$

$$\text{Charge 2} = 5 \times 10^{-6}$$

$$r = 3 \text{ cm} = 0.03 \text{ m}$$

$$\text{Current 1} = 2 \text{ A}$$

$$\text{Current 2} = 3 \text{ A}$$

$$L = 1 \text{ mm} = 0.001 \text{ m}$$

$$D = 10 \text{ m}$$

i) Force B/w Two Electric Charge

$$\text{Formula: } \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 \cdot q_2}{r^2}$$

$$\text{given, } q_1 = 3 \times 10^{-6} \text{ C}$$

$$q_2 = 5 \times 10^{-6} \text{ C}$$

$$r = 0.03 \text{ m}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$\text{Coulomb const } k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$$

Calculation:-

$$F = (8.99 \times 10^9) \times \frac{(3 \times 10^{-6})(5 \times 10^{-6})}{(0.03)^2}$$

$$F = (8.99 \times 10^9) \times \frac{15 \times 10^{-12}}{9 \times 10^{-4}}$$

$$F = (8.99 \times 10^9) \times (1.67 \times 10^{-8})$$

$$F = 150 \text{ N} //$$

$$\therefore 149.8 \text{ N} \approx 150 \text{ N}$$

Case 1 :

Given two point charges $q_1=2\times 10^{-6}$ and $q_2=3\times 10^{-6}$ C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m.

Explain how the force changes with the distance between the charges.

Code:

```
// Parameters for two point charges
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
q1 = 2e-6; // Charge 1 (Coulombs)
q2 = 3e-6; // Charge 2 (Coulombs)
r = linspace(0.1, 10, 100); // Distance between charges (m)

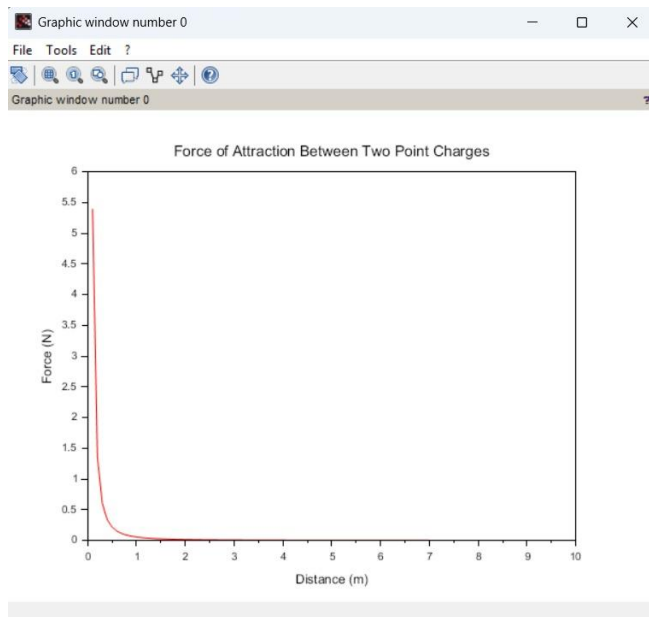
// Calculate force as a function of distance
F = ke * abs(q1 * q2) ./ r.^2;

// Plotting the force vs distance
plot(r, F, "r");
xlabel("Distance (m)");
ylabel("Force (N)"); // Parameters for two point charges
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
q1 = 2e-6; // Charge 1 (Coulombs)
q2 = 3e-6; // Charge 2 (Coulombs)
r = linspace(0.1, 10, 100); // Distance between charges (m)

// Calculate force as a function of distance
F = ke * abs(q1 * q2) ./ r.^2;

// Plotting the force vs distance
plot(r, F, "r");
xlabel("Distance (m)");
ylabel("Force (N)");
title("Force of Attraction Between Two Point Charges");
```

Output:



Case 2:

Given two point charges $q_1=5\times 10^{-6}$ C and $q_2=2\times 10^{-6}$, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.

Code:

```
// Parameters for two point charges
```

```
clc;
```

```
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
```

```
q1 = 5e-6; // Charge 1 (Coulombs)
```

```
q2 = 2e-6; // Charge 2 (Coulombs)
```

```
r = linspace(0.5, 5, 100); // Distance between charges (m)
```

```
// Calculate force as a function of distance
```

```
F = ke * abs(q1 * q2) ./ r.^2;
```

```
// Plotting the force vs distance
```

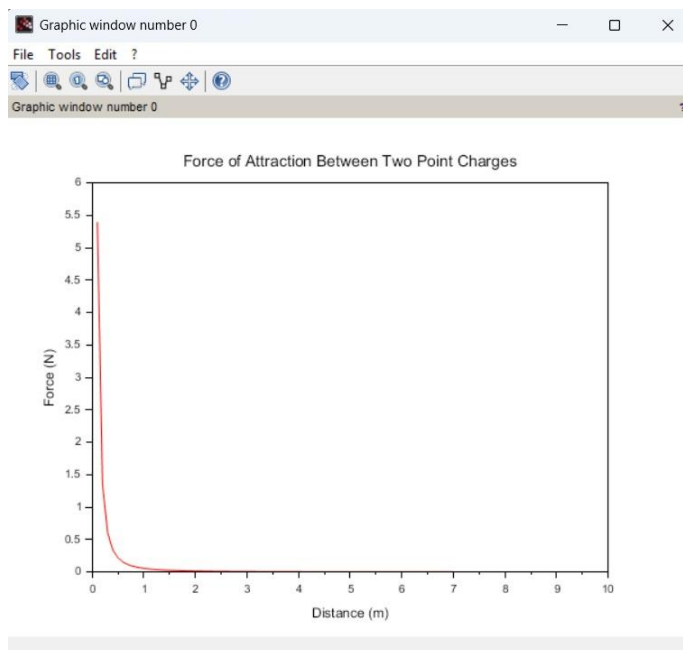
```
plot(r, F, "b");
```

```
xlabel("Distance (m)");
```

```
ylabel("Force (N)");
```

```
title("Force of Attraction Between Two Point Charges (Different Magnitudes)")
```

Output:



Case 3:

Two parallel conductors are separated by a distance d . The first conductor carries a current of $I_1=5$ A, and the second conductor carries a current of $I_2=10$ A. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors

Code:

```
// Parameters for two parallel conductors
```

```
clc;
```

```
mu0 = 4 * %pi * 1e-7; // Permeability of free space (N/A^2)
```

```
I1 = 5; // Current in conductor 1 (Amperes)
```

```

I2 = 10; // Current in conductor 2 (Amperes)

d = linspace(0.1, 10, 100); // Distance between conductors (m)

// Calculate force per unit length as a function of distance
F_per_length = (mu0 * I1 * I2) ./ (2 * %pi * d);

// Plotting the force per unit length vs distance
plot(d, F_per_length, "g");

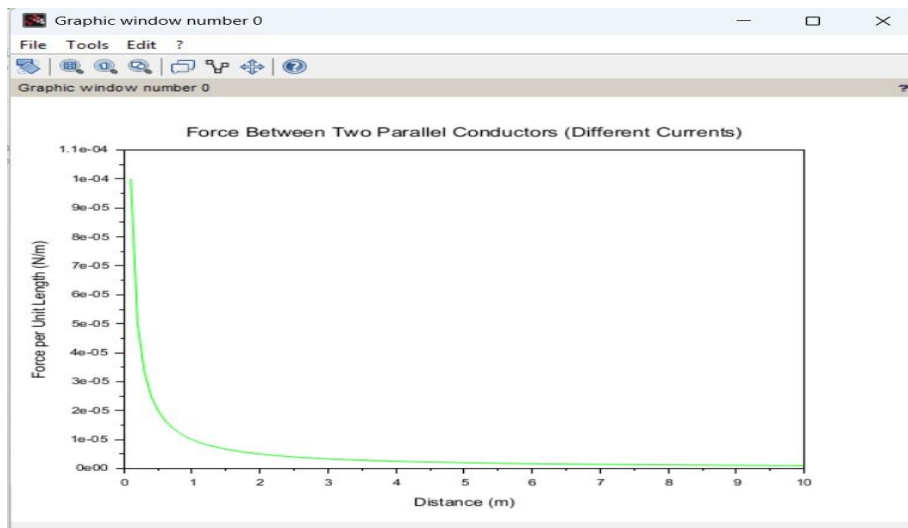
xlabel("Distance (m)");

ylabel("Force per Unit Length (N/m)");

title("Force Between Two Parallel Conductors (Different Currents)");

```

Output:



Case 4:

Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.

Code:

```

// Parameters for two parallel conductors

clc;

mu0 = 4 * %pi * 1e-7; // Permeability of free space (N/A^2)

I1 = 3; // Current in conductor 1 (Amperes)

I2 = 3; // Current in conductor 2 (Amperes)

d = linspace(0.1, 2, 100); // Distance between conductors (m)

```

```

// Calculate force per unit length as a function of distance
F_per_length = (mu0 * I1 * I2) ./ (2 * %pi * d);

// Plotting the force per unit length vs distance
plot(d, F_per_length, "m");

xlabel("Distance (m)");

ylabel("Force per Unit Length (N/m)");

title("Force Between Two Parallel Conductors (Same Current)");

```

Output:

